Understanding the physical properties that control protein crystallization by analysis of large-scale experimental data

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Crystallization is the most serious bottleneck in high-throughput protein-structure determination by diffraction methods. We have used data mining of the large-scale experimental results of the Northeast Structural Genomics Consortium and experimental folding studies to characterize the biophysical properties that control protein crystallization. This analysis leads to the conclusion that crystallization propensity depends primarily on the prevalence of well-ordered surface epitopes capable of mediating interprotein interactions and is not strongly influenced by overall thermodynamic stability. We identify specific sequence features that correlate with crystallization propensity and that can be used to estimate the crystallization probability of a given construct. Analyses of entire predicted proteomes demonstrate substantial differences in the amino acid–sequence properties of human versus eubacterial proteins, which likely reflect differences in biophysical properties, including crystallization propensity. Our thermodynamic measurements do not generally support previous claims regarding correlations between sequence properties and protein stability.

The field of structural genomics aims to capitalize on the recent proliferation of genomic sequence data by developing methods and infrastructure for high-throughput protein-structure determination using X-ray crystallography[1–5] and NMR. These efforts have contributed to improvements in expression and structure-determination methods[6]; indeed, the four largest US structural genomics consortia accounted for 45% of all novel structures deposited in the Protein Data Bank (PDB) in 2007 (ref. 7). Nevertheless, sequence information is accumulating much faster than structural information, a gap that will continue to expand in the absence of a breakthrough in methods for macromolecular-structure determination. Among structural genomics consortia, only ~35% of purified proteins form objects resembling crystals, and only ~12% form crystals of sufficient quality for structure determination, with exact frequencies varying based on protein source and the quality control imposed during sample production[7]. The vast majority of protein preparations do not yield crystal structures, which represents a severe technical obstacle in myriad projects. Drug-discovery efforts are particularly handicapped as lack of structural data precludes the use of structure-based ligand optimization methods[8,9].

In this study we investigated the physical properties that control protein crystallization using materials and results from the Northeast Structural Genomics Consortium (NESG). The availability of data on hundreds of biochemically well-behaved proteins produced and evaluated using consistent methods should reduce the influence of sporadic factors affecting crystallization outcome and allow sensitive detection of causally related properties. Experimental results and sequence parameters were correlated with whether a protein yielded a crystal of sufficient quality for determination of its atomic structure (that is, a PDB deposition), rather than simply with crystal formation.

To form a high-quality crystal, a protein must be immobilized in a lattice in a consistent conformation with limited dynamic motion. Thus, thermodynamic stability could play an important role in determining crystallization behavior. We evaluated this possibility using large-scale experimental studies of protein-folding equilibria. Protein surface properties could also play a determining role in controlling crystallization behavior because formation of tight, geometrically precise intermolecular contacts is required for lattice stability. Although this premise seems conceptually obvious[10,11], limited information has been available thus far concerning the specific features involved and whether such surface properties can be inferred from primary sequence.

Some studies have considered possible relationships between protein surface properties and crystallization propensity. Conformationally dynamic amino acid side chains may inhibit crystallization because of the entropic cost of immobilizing them in stable

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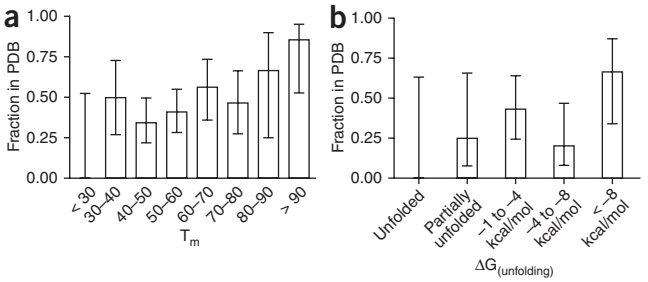

Figure 1 Protein stability does not strongly influence success in crystal-structure solution. Denaturation experiments were performed on biochemically well-behaved NESG proteins drawn from the set of 679 used in the data mining studies (as described in detail in **Supplementary Methods**). The graphs show the fraction of proteins for which crystal structures were successfully determined and deposited in PDB in each stability bin. The error bars represent 95% confidence limits calculated from counting statistics using the numbers in each bin. (**a**) Crystallization success versus thermal denaturation midpoint temperature (T_m) determined by fluorescence enhancement of the hydrophobic reporter dye SYPRO Orange[23]. Logistic regression analyses of the probability of the observed relationships occurring at random indicate $P = 0.0076$ for all proteins ($N = 117$), $P = 0.19$ for those with $T_m \leq 90$ °C ($N = 110$), $P = 0.020$ for those with $T_m \geq 30$ °C ($N = 114$) and $P = 0.40$ for those with T_ms between 30 °C and 90 °C ($N = 107$). (**b**) Crystallization success versus ΔG of unfolding from guanidinium hydrochloride denaturation experiments monitored by circular dichrosism spectroscopy ($P = 0.2$, $N = 36$ for all proteins).

interprotein contacts. 'Surface entropy reduction' by replacing the high-entropy side chains of surface-exposed lysine, glutamate and glutamine residues with lower-entropy residues, especially alanine[12–17], has promoted crystallization of some proteins, although such substitutions may also decrease protein solubility, hindering crystallization screening[14,17]. In addition, several reports have described significant correlations between sequence properties and crystallization propensity[18–22], with low average hydrophobicity[18,21], high isoelectric point (pI)[18,21] and long stretches of disordered backbone[18,19] reducing crystallization probability. However, these studies have not addressed the mechanistic origin of the observed correlations.

We present evidence that the prevalence of low-entropy, well-ordered surface features is the principal mechanistic determinant of protein crystallization behavior. Our analyses suggest approaches to rational mutational reengineering of proteins to improve crystallization propensity by increasing the prevalence of residues that positively correlate with successful crystal-structure determination. Our data also permit evaluation of relationships between various biophysical and sequence properties of proteins, allowing critical evaluation of trends previously proposed but not rigorously tested.

RESULTS

Does protein stability influence crystallization?

We used several biophysical techniques to evaluate protein stability and assess its relationship to successful crystal-structure determination. Thermal denaturation experiments were carried out on 117 monodisperse proteins that had gone through the NESG expression and crystallization pipeline. In these experiments, melting temperature (T_m) was determined by means of the fluorescence of the dye SYPRO Orange, which is enhanced upon partitioning into the hydrophobic regions of denatured proteins[23] (**Fig. 1a**). Chemical denaturation of 36 proteins by guanidinium hydrochloride was monitored using circular dichroism spectroscopy (**Fig. 1b**). Finally, 121 proteins were subjected to calibrated limited proteolysis using trypsin and proteinase K (**Supplementary Fig. 1** online). Results from these three assays are consistent (**Supplementary Fig. 2** and **Supplementary Table 1** online) and show no significant relationship between overall stability and structure determination if unfolded and hyperstable proteins are omitted (**Fig. 1** and **Supplementary Fig. 1**). Thermal denaturation data (**Fig. 1a**) show a statistically significant correlation if all proteins are included in the analysis ($P = 0.008$), but the significance is lost if proteins with T_m's <30 °C or >90 °C are excluded ($P = 0.4$). Therefore, partially or fully unfolded proteins may yield crystal structures less frequently, whereas hyperstable proteins probably yield crystal structures somewhat more frequently. However, overall thermodynamic stability is not a major determinant of crystallization propensity and may not have any influence across the broad range of stabilities typical of folded mesophilic proteins (see **Supplementary Notes** online for further discussion).

A detailed discussion of the large-scale proteolysis results is presented in **Supplementary Notes**. Most notably, the size of the dominant protected fragment in proteolysis studies, which is likely to be a measure of the total content of disordered loops, significantly ($P = 0.015$) positively correlates with successful crystal-structure determination (**Supplementary Fig. 1**). Our large-scale experimental analyses also allow evaluation of the relationship between thermodynamic and sequence properties of biochemically well-behaved proteins (**Supplementary Notes**, **Supplementary Figs. 2** and **3** and **Supplementary Table 1** online). Our data on primarily bacterial proteins do not support the conclusion[24] that specific combinations of hydrophobicity and net charge reliably identify natively unfolded proteins (**Supplementary Fig. 4** online). Finally, our data show an insignificant correlation between fluorescence enhancement of the hydrophobic reporter dye *bis*-ANS and either the folding state or the crystallization propensity of a set of NESG proteins (**Supplementary Notes** and **Supplementary Fig. 5** online).

Influence of other biophysical properties

The hydrodynamic properties of all proteins studied by the NESG are characterized using analytical gel filtration chromatography monitored by static light scattering and refractive index detectors. These data, acquired from a flash-frozen aliquot of the crystallization stock, provide a rigorous description of the distribution of oligomeric species in each sample. Theoretical considerations suggest that protein oligomers should crystallize more readily than monomers because a single high-quality packing epitope on the surface of one protomer can make repeated contacts, which is especially beneficial in a lattice sharing the symmetry of the oligomer[25]. This inference is supported by a study in which disulfide cross-linking was used to produce non-native dimers, which crystallized more avidly than the corresponding monomers[26]. Our statistical analysis of NESG large-scale crystallization results proves this inference, showing that monomers yield solvable crystals at a significantly lower rate compared with dimers ($P = 0.0005$) or larger oligomers ($P = 0.0002$) (**Fig. 2a**). We also found that the hydrodynamic homogeneity of the protein stock significantly correlates with successful crystal-structure determination (**Fig. 2b**). Monodisperse proteins ($\geq$90% in a single hydrodynamic species) yield crystal structures more frequently than do predominantly monodisperse (70–90%) ($P = 0.01$) or polydisperse ($<$70%) ($P = 0.03$) proteins. The few aggregated proteins that entered the NESG pipeline during the period analyzed here failed to yield structures (**Fig. 2b**). Preliminary analysis of results from a substantially larger dataset containing $>$100 aggregated proteins shows the same trend (data not shown). Therefore, although formation of specific, homogeneous

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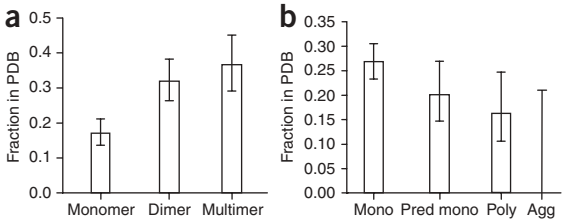

Figure 2 Hydrodynamic properties strongly influence success in crystal structure solution. The graphs show the fraction of proteins for which crystal structures were successfully determined and deposited in PDB, with the error bars representing 95% confidence limits calculated from counting statistics using the numbers in each bin. (**a**) Crystallization success versus oligomeric state ($P = 0.0005$ for monomers compared to dimers, $P = 0.0002$ for monomers compared to larger multimers and $P = 0.000007$ for monomers compared to all oligomers). (**b**) Crystallization success versus aggregation status ($P = 0.01$ for monodisperse (Mono) compared to all other classes and $P = 0.03$ for at least predominantly monodisperse (Pred mono) compared to polydisperse (Poly) and aggregated (Agg)).

oligomers promotes successful crystallization (**Fig. 2a**), heterogeneous self-association inhibits successful crystallization (**Fig. 2b**).

Finally, we determined that the number of unvalidated crystallization hits observed in a 1,536-well robotic high-throughput microbatch screen[27] correlates strongly ($P = 8.4 \times 10^{-19}$) with crystal-structure solution (**Supplementary Fig. 6** online). We hypothesize that proteins crystallizing more promiscuously possess more interprotein interaction sites on their surfaces, or more strongly interacting sites, and that at least one of these properties is a significant determinant of the ability to form a high-quality crystal lattice under some conditions. The fact that structures were determined for only ~6% of the proteins that failed to give a hit in initial high-throughput screening, despite being tested in ≥500 additional vapor diffusion reactions, shows that crystallization propensity is an intrinsic protein property even though it can be influenced by solution composition (**Supplementary Notes**).

Data mining historical crystallization results

We examined the relationship between the sequence properties of NESG proteins and success in crystal structure determination. The dataset comprised 679 strongly expressed and well-behaved proteins, predominantly but not entirely from bacterial organisms, of which 157 yielded crystal structures. These proteins were taken through the entire NESG pipeline[6], including quality-control assays and crystallization screening. Proteins with predicted transmembrane α-helices or >20% low-complexity sequence were excluded from the pipeline. The dataset included just one construct for each protein target and excluded proteins identified by static light scattering as aggregated in their crystallization stocks. Samples yielding crystals of insufficient quality for structure determination were considered failures even if diffraction was observed, which occurred in 39 of 679 cases. Analyses retrospectively justify this strategy by showing that some key sequence features in these 39 proteins are more similar to those of proteins that failed to yield crystal structures than to those of proteins that did (**Supplementary Fig. 7** online). Target selection and classification are described in **Supplementary Methods** online.

Sequence properties that were analyzed include the frequency of each amino acid, mean hydrophobicity or grand average of hydropathy (GRAVY[28]), mean side-chain entropy[29] (⟨SCE⟩), total and net electrostatic charge, pI, the fraction of residues predicted to be disordered by DISOPRED2[30] and chain length. Logistic regressions, explained briefly in the **Supplementary Notes**, were performed to evaluate the dependence between the continuous variable representing the sequence parameter and the binary outcome of the crystallization effort, namely success or failure in depositing the crystal structure into the PDB. The *P*-value, regression slope and predictive value of each variable are presented in **Figure 3** and **Table 1a**.

The frequencies of five amino acids significantly correlate with successful crystal-structure determination, as do several more complex sequence metrics. Alanine, glycine and phenylalanine frequencies positively correlate ($P = 0.0019$, 0.0046 and 0.014, respectively) with successful structure determination, whereas glutamic acid and lysine frequencies negatively correlate ($P = 0.0029$ and 0.0018, respectively) (**Fig. 4** and **Supplementary Fig. 8** online). GRAVY positively correlates ($P = 0.0000135$), whereas ⟨SCE⟩ and the fraction of predicted disordered residues negatively correlate ($P = 0.000000915$; **Fig. 4**), as do fractional positive ($P = 0.000665$; arginine + lysine) or negative ($P = 0.0144$; asparagine + glutamic acid) charge (**Supplementary Fig. 9** and **Supplementary Table 2** online). Fractional values (that is, normalized to chain length) are uniformly more predictive than total values for significantly correlated parameters (**Supplementary Fig. 10** online). Although fractional positive or negative charge considered independently both significantly ($P = 0.000665$ and 0.0144, respectively) oppose successful crystallization, neither net charge (positive minus negative residue counts) nor its absolute value are predictive, nor are any electrostatic variables without length normalization (**Supplementary Table 2** and **Supplementary Fig. 9**). Note that the current analysis has a limited sensitivity in detecting the influence of rare or weakly predictive amino acids due to the size of our dataset (**Supplementary Notes**).

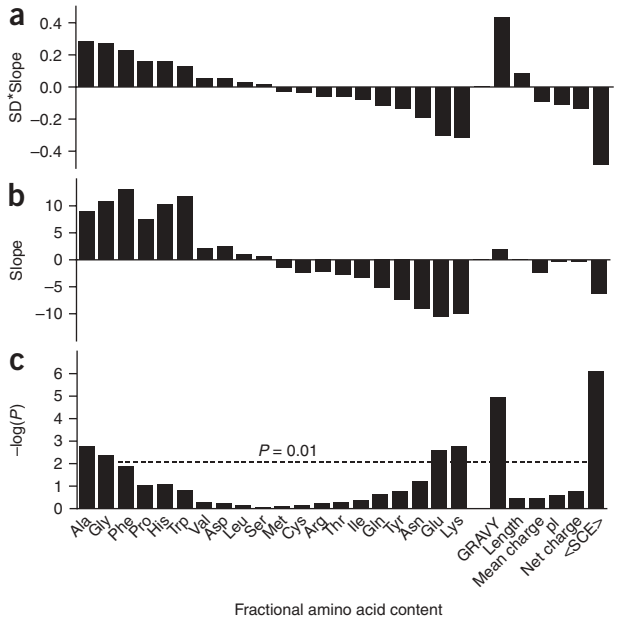

Figure 3 Correlations between sequence characteristics and success in crystal structure solution. Logistic regressions based on success in crystal structure determination (that is, PDB deposition) were performed on a dataset comprising 679 proteins from the NESG protein expression and crystallization pipeline. Variables evaluated included the fractional content of each amino acid, mean residue hydrophobicity (GRAVY[28]), chain length, mean charge (fraction arginine+lysine+asparagine+glutamatic acid), pI, mean net charge and ⟨SCE⟩. (**a**) Predictive value of each parameter, which is defined as the product of its logistic regression slope and the s.d. of its distribution in the dataset. (**b**) Logistic regression slope. (**c**) Negative log of logistic regression *P*-value.

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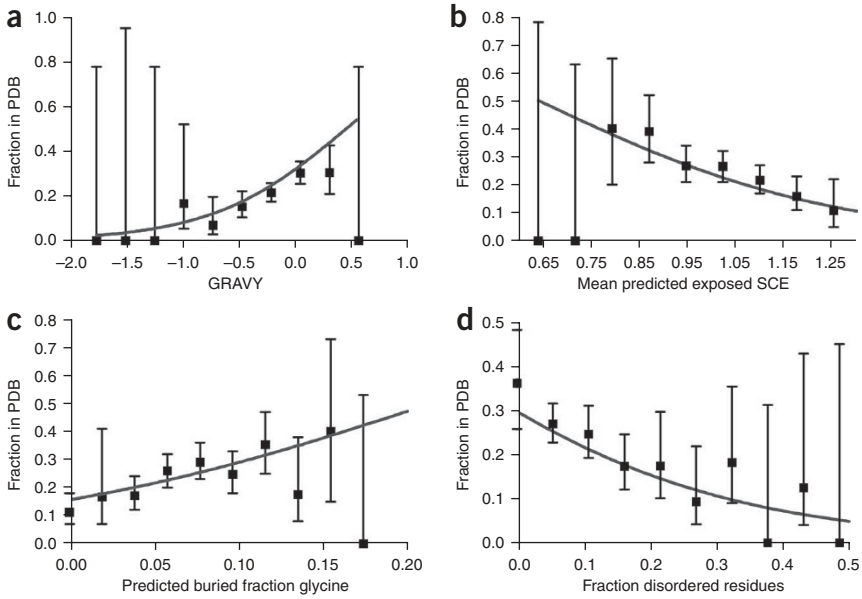

Figure 4 Four major predictors of success in crystal structure solution. Graphs show the fraction of 679 NESG pipeline proteins for which crystal structures were successfully determined and deposited in PDB. Error bars represent 95% confidence limits calculated from counting statistics using the numbers in each bin, whereas gray lines show the functional forms of the optimized logistic regression equations. (**a**) GRAVY[28] or mean residue hydrophobicity, a positive predictor ($P = 0.0000135$). (**b**) $\langle$SCE$\rangle$[33] of PHD/PROF[32] predicted exposed residues, a negative predictor ($P = 0.0001$). (**c**) Fractional glycine content of PHD/PROF[32] predicted buried residues, a positive predictor ($P = 0.0005$). (**d**) Fraction of residues predicted to be disordered by DISOPRED2[30], a negative predictor ($P = 0.0003$).

Protein pI (**Supplementary Fig. 11** online) and chain length (**Supplementary Fig. 12** online) both show bimodal effects, with the rate of success initially increasing and later decreasing with increasing parameter values. The bimodal dependence prevents assessment of statistical significance using logistic regression, and the 95% confidence limits of all parameter bins substantially overlap in our dataset. However, similar dependencies have been reported in independent datasets[18–22] and can be observed in analyses of whole genome distributions (see below), suggesting that these parameters influence crystallization outcome but not strongly enough to be significant in our dataset.

Predicted backbone disorder inhibits crystallization

We further analyzed the effect of backbone disorder in opposing successful crystallization (**Fig. 4**) by performing multiple logistic regression on the fraction of predicted disordered residues[30] located in continuous segments at either the N or C terminus of the protein or at internal positions (summed together). The regression slopes in **Table 1b** demonstrate that predicted disorder has the same size effect opposing crystallization irrespective of location within the protein chain.

Surface entropy effects dominate other effects

GRAVY and $\langle$SCE$\rangle$, which both strongly influence crystallization outcome (**Figs. 3** and **4** and **Table 1a**), are anti-correlated (**Supplementary Table 3** online). Analyzing them simultaneously in a double logistic regression demonstrates that their influence is redundant: GRAVY shows weak and insignificant additional correlation with crystallization outcome when considered simultaneously with $\langle$SCE$\rangle$ (**Table 1c**). The statistical dominance of $\langle$SCE$\rangle$ over GRAVY indicates that it is more strongly correlated and therefore likely to be the mechanistically significant parameter. This evidence supports the hypothesis that the thermodynamic cost of immobilizing high-entropy side chains tends to inhibit their participation in crystal-packing contacts[15,16,31]. As their prevalence on the protein surface increases and fewer well-ordered sites are available for packing, the probability increases that thermodynamically unfavorable immobilization of high side-chain entropy residues will be needed to form a stable contact[15,16]. The comparatively weak anti-correlation of some high side-chain entropy residues (e.g., arginine in **Fig. 3**) may be attributable to an energetically favorable interaction tendency of the side chain's functional group partially offsetting the entropic cost of immobilizing it (**Supplementary Notes**).

This mechanistic hypothesis implies that residues influencing crystallization should be localized on the protein surface. Therefore, we segregated the amino acids in each protein sequence by their predicted location (buried versus surface-exposed) according to PHD/PROF[32], a secondary structure prediction program suite (**Table 1a**). This analysis strongly supports the hypothesis, showing that successful crystallization correlates with $\langle$SCE$\rangle$ in predicted exposed residues ($\langle$SCE$\rangle_{pe}$) but not predicted buried residues. In contrast, the correlation with GRAVY is inconsistent in direction and substantially weaker when residues are segregated by predicted location. The direction of the GRAVY correlation in exposed residues (**Table 1a**) is consistent with it being a surrogate for $\langle$SCE$\rangle$ because higher hydrophobicity correlates with lower side-chain entropy (**Supplementary Table 3**). Furthermore, when sets of proteins are assembled with equivalent $\langle$SCE$\rangle$ distributions but systematic differences in GRAVY, they show no significant differences in crystal structure–determination rates (**Supplementary Fig. 13** online). These results all reinforce the conclusion that increasing exposed side-chain entropy, rather than decreasing hydrophobicity, impedes successful crystal-structure determination by inhibiting formation of stable packing contacts.

Like GRAVY, other sequence properties observed to have a significant influence on crystal structure determination outcome strongly correlate with side-chain entropy. Alanine and glycine have the lowest side-chain entropies, and their frequencies correlate most strongly with successful crystallization, whereas lysine and glutamic acid have among the highest side-chain entropies, and their frequencies anti-correlate most strongly. Multiple logistic regression analyses suggest that all fractional charge effects (**Supplementary Table 2**) and all single amino acid effects are redundant with $\langle$SCE$\rangle_{pe}$ except for the frequencies of predicted buried glycine and exposed phenylalanine, which remain significant when considered simultaneously with $\langle$SCE$\rangle_{pe}$ (**Table 1d**). Furthermore, in sets of proteins assembled to have equivalent $\langle$SCE$\rangle_{pe}$ distributions but systematic differences in individual amino acid frequencies, higher fractional content of glycine, alanine or phenylalanine significantly ($P = 0.000069$, 0.0012 and 0.038, respectively) increases crystal structure–determination rate whereas higher fractional content of lysine, glutamic acid or charged residues does not significantly alter outcome (**Supplementary Fig. 13**). These

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Table 1 Single and multiple logistic regression results[a]

	Regression variable	Slope	S.D.*Slope	*P* value
	GRAVY	1.68	0.43	**0.0000135**
	⟨SCE⟩	–5.99	–0.47	**0.000000915**
	Gly	10.685	0.26	**0.0046**
	Pb ⟨SCE⟩	–1.15	–0.072	0.431
a	**Pe ⟨SCE⟩**	–3.24	–0.33	**0.0001**
	Pb GRAVY	–0.413	–0.25	**0.0085**
	Pe GRAVY	0.744	0.26	**0.0044**
	Pb Gly	9.16	0.32	**0.0005**
	Pe Gly	3.03	0.089	0.08
	DISOPRED2	–4.21	–0.46	**0.0003**
	N/C/Internal DISOPRED2			**0.0009**
b	N-terminal DISOPRED2	–4.37	–0.18	0.15
	C-terminal DISOPRED2	–4.14	–0.21	0.093
	Internal DISOPRED2	–4.19	–0.31	**0.013**
	⟨SCE⟩ + GRAVY			**0.00000181**
c	⟨SCE⟩	–4.52	–0.36	**0.0066**
	GRAVY	0.73	0.19	0.16
	Pb/e ⟨SCE⟩ + amino acids			**0.000085**
	Pb ⟨SCE⟩	2.68	0.17	0.20
	Pe ⟨SCE⟩	–3.66	–0.40	**0.038**
	Pb Ala	3.68	0.17	**0.015**
	Pe Ala	2.89	0.11	**0.036**
	Pb Glu	3.86	0.037	0.72
d	Pe Glu	–0.0186	–0.00080	0.99
	Pb Gly	12.1	0.42	**0.00020**
	Pe Gly	–1.51	–0.045	0.71
	Pb Lys	–14.7	–0.10	0.38
	Pe Lys	2.0	0.096	0.50
	Pb Phe	3.71	0.17	0.10
	Pe Phe	19.5	0.22	**0.02**

[a]Logistic regressions evaluating success in depositing a crystal structure in the PDB as a function of various sequence parameters were performed on a set of 679 biochemically well-behaved NESG proteins (157 of which yielded structures). For multiple logistic regressions, parameters for the individual component variables are shown in plain type whereas the overall results are shown in bold type. All *P*-values <0.05 are shown in bold type. The 'predictive value' is the product of the regression slope for each variable times the s.d. of its distribution in the dataset. GRAVY represents the GRand AVerage of hydropathY[28], ⟨SCE⟩ mean Monte Carlo side-chain entropy[33] and DISOPRED2[30] the fraction of backbone disorder predicted by this program (at a false-positive rate of 5%). Pb and Pe refer to the fraction of residues predicted to be buried or surface-exposed, respectively, by the PHD/PROF algorithm as implemented by the PredictProtein server[32]. N-terminal and C-terminal refer to continuous stretches of amino acids at the ends of the protein, and internal refers to all internal positions combined.

results suggest that effects of the latter parameters are proxies for the mechanistically dominant effect of ⟨SCE⟩, whereas glycine, alanine and phenylalanine may have mechanistically independent positive effects. These residues may preferentially mediate crystal-packing contacts. The significant effect of fractional alanine content in the analysis of sets of proteins with equivalent $\langle SCE \rangle_{pe}$ distributions (**Supplementary Fig. 13**) but not in multiple logistic regression (**Table 1d**) is likely to be attributable to the fact that its influence on crystallization rate does not match the logistic functional form as well as those of glycine or phenylalanine (**Supplementary Fig. 8**).

The 'buried' glycine effect

Although the glycines that promote crystallization are predicted by PHD/PROF to be buried, the same program also predicts them to be preferentially localized in loops that are 6–15 residues in length (**Supplementary Table 4** online). The **Supplementary Notes** present detailed analyses of this ostensibly inconsistent categorization. In brief, manual inspection of crystal structures suggests that the predicted 'buried' loop category is dominated by well-ordered glycine residues partially exposed on the surface of the protein, which may be favorable sites to form crystal-packing contacts (**Supplementary Fig. 14** online).

P_{XS}: probability of crystal-structure determination

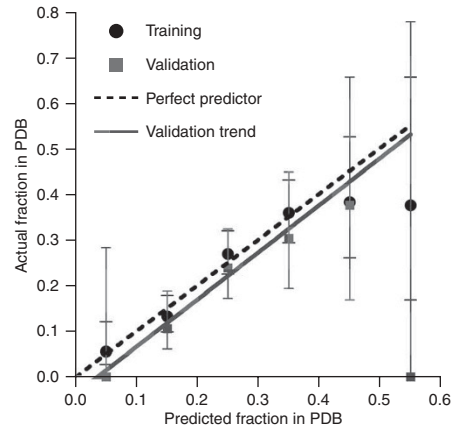
Having identified four nonredundant sequence features showing statistically significant correlation with crystallization success, we combined these into a single predictive metric (**Supplementary Fig. 15** online):

$$P_{XS} = 1/(1+e^{-(1.85-3.20*\mathrm{Diso}-3.77*\langle SCE\rangle_{pe}+8.14*G_{pb}+14.26*F)}).$$

P_{XS} represents the probability of solving a crystal (xtal) structure, Diso the fraction of residues predicted to be disordered by DISOPRED2, $\langle SCE \rangle_{pe}$ the mean side-chain entropy of predicted exposed residues, G_{pb} the fraction of predicted buried glycine and F the fraction of total phenylalanine (used because this parameter predicts more strongly than exposed phenylalanine alone). This metric provides an accurate description of the behavior of the training dataset up to bins with ~35% success in depositing a crystal structure ($P = 5.3 \times 10^{-9}$) (**Fig. 5**). More importantly, it provides a similarly accurate description of a validation dataset comprising 200 proteins that passed through the NESG pipeline after metric development ($P = 0.0014$). A web server performing this calculation is available at http://www.nesg.org/PXS/.

Genome-wide analyses of crystallization propensity

In addition to possible bias from inconsistent methods and effort being applied to different proteins, genome-wide crystallization results



Bin Center	0.05	0.15	0.25	0.35	0.45	0.55	Total
N (Train.)	72	196	238	131	34	8	679
N (Valid.)	8	66	84	33	8	1	200

Figure 5 Performance of the P_{XS} metric predicting probability of successful crystal-structure determination. Box and whisker plot shows fraction of NESG pipeline proteins in PDB with 95% confidence limits calculated from counting statistics using the numbers in each bin, binned as a function of P_{XS} value. Black represents the 679 proteins used to develop or train the metric, whereas gray represents 200 proteins produced at later dates that were used for validation. The solid lines represent the functional form of the logistic regression equation describing P_{XS} as applied to the sequences in training (dotted black) or validation (solid gray) sets. The table shows the number of proteins falling into each bin on the graph for the training (Train.) and validation (Valid.) sets.

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are systematically influenced by protein expression and solubility characteristics, factors intentionally excluded from the analyses reported above so as to isolate parameters influencing crystallization. To begin characterizing the interplay of potentially conflicting factors influencing the successive steps required to go from gene to protein structure, and to explore the generality of P_{XS}, we analyzed proteome-wide distributions of its value and the underlying sequence parameters (**Supplementary Notes**). In brief, P_{XS} is significantly predictive of the crystal structures obtained from the human and *Escherichia coli* proteomes, as are all individual sequence parameters predictive in the NESG dataset, except for $\langle SCE \rangle_{pe}$ (**Supplementary Figs. 16**, **17** and **18** online). Sequence parameter distributions differ dramatically between *E. coli* and human proteins, which have on average more backbone disorder, lower GRAVY and lower $\langle SCE \rangle_{pe}$ (**Supplementary Fig. 17**). Human but not *E. coli* proteins have a very high prevalence of low side-chain entropy residues, especially glycine and proline, in disordered sequences (**Supplementary Fig. 19** online). The **Supplementary Notes** also present a metric developed to predict the conflated probability of expressing and determining the crystal structure of a human protein ($P_{C\text{-}XS\text{-}Hs}$; **Supplementary Fig. 20** online).

DISCUSSION

Our statistical analysis of large-scale protein crystallization results demonstrates that the mean entropy of exposed side chains and predicted backbone disorder both anti-correlate strongly and significantly with successful structure determination. Combining these results with the observation that stability is not a significant determinant of success leads to the conclusion that the dominant factor determining protein crystallization outcome is the prevalence of well-ordered surface epitopes capable of mediating stereochemically specific interprotein packing interactions. Beyond offering rigorous confirmation of longstanding suspicions[12–16], our results provide a quantitative metric to assess crystallization propensity (P_{XS}) and suggest possibilities for engineering protein sequences to improve outcome. Whereas previously reported sequence properties that correlate with crystallization propensity[18–22] appear to be surrogates for surface entropy, we find that the frequencies of glycine, alanine and phenylalanine have statistically significant independent effects on crystal structure determination success. We hypothesize that these residues are particularly effective in mediating crystal-packing interactions, presumably by means of amide backbone interactions for glycine and alanine and hydrophobic side-chain interactions for phenylalanine. Our experimental studies demonstrate that heterogeneous self-association, which produces polydispersity or aggregation in a stock solution, significantly reduces crystallization probability. Thus, successful crystallization requires minimal protein self-association in dilute aqueous buffers but strong protein self-association under the low water-activity conditions used to form a crystal, which is a nonphysiological protein aggregate, albeit one with consistent intermolecular contacts and spatial organization. These requirements fundamentally tend to conflict. The charged residues lysine and glutamic acid promote solubility but impede crystallization (**Fig. 3**), whereas low solubility and crystallization are both driven by low-affinity, nonphysiological intermolecular interactions. Optimal crystallization epitopes should mediate strong stereospecific interactions under low water-activity conditions without promoting promiscuous surface interaction in dilute aqueous buffers. Well-ordered, surface-exposed glycines may be particularly efficacious in this regard compared to residues with stronger hydrophobic character, which tend to promote nonspecific interactions (see **Supplementary Notes** for a more detailed discussion of related issues).

METHODS

Protein crystallization. Protein expression, purification and analysis methods are described in **Supplementary Methods**. Initial high-throughput crystallization screening was conducted using the 1,536-well microbatch robotic screen at the Hauptmann-Woodward Institute[27]. Proteins failing to yield rapidly progressing crystal leads were subjected to vapor diffusion screening, typically 300–500 conditions (Crystal Screens I & II, PEG-Ion and Index screens from Hampton Research or equivalent screens from Qiagen) at both 4 °C and 20 °C. Screening was conducted in the presence of substrate or product compounds if commercially available. Crystal optimization, diffraction data collection at cryogenic temperatures, structure solution using single or multiple-wavelength anomalous diffraction techniques and refinement were conducted using standard methods.

Data mining methods. Data mining analyses were conducted on native sequences with tags removed. Hydrodynamic data from SPINE (Structural Proteomics in the NorthEast)[33,34] were manually verified. The frequency of each amino acid and the compound sequence metrics of charge, pI, GRAVY, SCE, length and DISOPRED[30] were individually evaluated for correlation with the binary outcome of success or failure in depositing a crystal structure of the target protein into the PDB. Charge parameters were calculated as signed or unsigned sums of the frequencies of appropriate combinations of arginine, lysine, glutamate and aspartate residues. Isoelectric point was calculated using the EMBOSS algorithm[35] at ExPASy[36]. GRAVY was calculated using the Kyte-Doolittle hydropathy parameters[28]. The Creamer scale was used for the side-chain entropy residue values of the individual amino acids[29]. DISOPRED scores were calculated using a locally installed copy of the DISOPRED2[30] program with a 5% false-positive rate. Calculations of predicted burial or exposure and secondary structure were performed with the PHD/PROF algorithms from the PredictProtein server[32,37]. Mean predicted exposed SCE was calculated as the mean for all residues predicted to be exposed, whereas all calculations based on secondary-structure class used total chain length as the denominator. In data graphs, the observed frequencies of successful PDB deposition in equally spaced parameter bins on the abscissa are plotted at the bin center, except for the terminal bin of unbounded variables, which is plotted at the average parameter value in the bin (**Supplementary Datasets 1–3**).

Statistical analyses. Logistic regressions were performed in STATA (Statacorp) with significance determined from Z-scores for individual variables and chi-squared distributions for models. The significance of oligomeric state, aggregation state and the dividing line in the charge/hydrophobicity chart (**Supplementary Fig. 4**) were determined by evaluating contingency tables with a two-tailed Fisher's exact test. Counting-statistics-based 95% confidence intervals were calculated using integration of the binomial distribution.

Note: Supplementary information is available on the Nature Biotechnology website.

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